Notes: Amiti & Konings (AER, 2007)

Trade Liberalization, Intermediate Inputs, and Productivity: Evidence from Indonesia

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1 Big picture

- **Question.** What are the productivity effects of reducing output tariffs versus input tariffs, and do firms that directly import intermediate inputs benefit more from tariff cuts than firms that do not import?
- **Setting.** The paper studies Indonesian manufacturing plants from 1991 to 2001 using the annual census of all firms with 20 or more employees. A key feature of the data is plant-level information on imported inputs and very detailed information on input use.
- **Core empirical challenge.** A fall in final-goods tariffs and a fall in input tariffs happen together, so one can easily misattribute the source of productivity gains. Measured productivity may also reflect markups, product switching, exchange-rate movements during the Asian crisis, and selection into importing or exporting.
- **Main conceptual contribution.** The paper separates the *import-competition* effect of lower output tariffs from the *input-access* effect of lower input tariffs. It also distinguishes direct gains to importing firms from possible spillovers to nonimporters.
- **Main hypotheses.**
  - Lower output tariffs should raise productivity by intensifying import competition.
  - Lower input tariffs should raise productivity by making foreign inputs cheaper and more accessible.
  - If imported inputs matter through embodied technology, better quality, or more varieties, the gains from lower input tariffs should be strongest for firms that actually import.
  - If importers become more efficient, some smaller indirect benefits may spill over to nonimporting firms through domestic competition or vertical linkages.
- **Main empirical design.** The authors first estimate plant-level TFP from a Cobb–Douglas production function using a modified Olley–Pakes procedure that treats importing and exporting decisions as additional state variables. They then regress TFP on output tariffs, input tariffs, and input tariffs interacted with importer status, using firm fixed effects and island-year fixed effects.
- **Main baseline result.** Once input tariffs are included, the productivity effect of output-tariff reductions becomes much smaller. In the baseline, a 10 percentage point fall in output tariffs raises productivity by only about **0.7%**, while a 10 percentage point fall in input tariffs raises productivity by about **12%** for importing firms and about **3%** for nonimporters.
- **Key heterogeneity result.** The benefits from lower output tariffs appear only in relatively competitive industries, while the gains from lower input tariffs for importers are present in both concentrated and competitive industries.
- **Key credibility result.** The main pattern survives alternative productivity measures, alternative tariff aggregation schemes, exclusion of foreign firms or licensed importers, product-switching controls, exchange-rate controls, Asian-crisis controls, alternative differencing estimators, and IV estimates.

- **Main economic interpretation.** The evidence is most consistent with large direct productivity gains from imported intermediate inputs—through foreign technology embodied in inputs, higher quality, more varieties, or learning—and with the standard import-competition channel being present but much smaller.
- **Economic takeaway.** In a liberalizing developing economy, tariff reductions on *inputs* can matter more for productivity than tariff reductions on *outputs*. The firm-level ability to import is central for understanding who benefits most.

2 Data and measurement (institutional context and key objects)

2.1 Why Indonesia is the right laboratory

- Indonesia experienced broad tariff reductions during the 1990s, especially after its 1995 WTO commitments.
- Tariffs were highly dispersed across the production chain and typically exhibited escalation: lower tariffs on inputs and higher tariffs on final goods.
- The data allow the authors to observe not only firm output and inputs, but also whether firms import inputs, export output, and have foreign ownership.

Interpretation. Indonesia provides unusually rich plant-level data in a period of sharp, highly disaggregated trade reform.

2.2 Manufacturing survey and sample construction

- The main source is the *Survei Industri* (SI), an annual census of Indonesian manufacturing firms with at least 20 employees.
- The sample covers **1991–2001**.
- The data include output, employment, intermediate inputs, capital, imports, exports, and foreign ownership.
- After cleaning for missing variables and unrealistic spikes, the final dataset is an unbalanced panel with about **15,000 firms per year** and **170,741 observations** in total.

Interpretation. The paper is built on a near-universe of formal manufacturing plants rather than a small survey sample.

2.3 Output tariffs

For each five-digit ISIC industry k, the output tariff is constructed as a simple average of HS nine-digit tariff lines mapped into that industry:

$$\text{output tariff}_{kt} = \frac{1}{N_k} \sum_{h \in k} \tau_{ht}. \tag{1}$$

Interpretation. This is the tariff protecting the firm's own output market, so its effect captures import competition.

2.4 Input tariffs

The key input-tariff measure is a weighted average of output tariffs on all inputs used by an industry:

$$\text{input tariff}_{kt} = \sum_j w_{jk}^{1998} \, \text{output tariff}_{jt}, \tag{2}$$

where

$$w_{jk}^{1998} = \frac{\sum_i \text{input}_{ijk}^{1998}}{\sum_{i,j} \text{input}_{ijk}^{1998}}. \tag{3}$$

- The weights come from plant-level input-use data in **1998**, the only year with detailed input composition.
- The input tariff is constructed at the **industry** level, not the firm level, to avoid endogeneity and sample-selection problems.
- The weights are based on *total* input use, including domestically purchased inputs, not just imported inputs.

Interpretation. This construction asks: given the typical input mix of an industry, how much tariff protection is embedded in the inputs it uses?

2.5 Why the input-tariff construction matters

- The plant-level input information is unusually detailed: firms list each intermediate input and the amount spent on it in rupiah.
- The 1998 plant data are aggregated into a **288-industry** manufacturing input-output table.
- Using total input shares rather than imported-input shares avoids mechanically assigning lower tariffs to firms that already import efficiently.

Interpretation. The paper tries to assign the economically relevant tariff exposure to each downstream industry rather than merely the tariff paid by current importers.

2.6 Productivity measurement

The starting production function is:

$$Y_{it} = A_{it} L_{it}^{\beta_l} K_{it}^{\beta_k} M_{it}^{\beta_m}. \tag{4}$$

In logs:

$$y_{it} = \beta_0 + \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + e_{it}. \tag{5}$$

Measured TFP is then:

$$\text{tfp}_{it}^{k} = y_{it} - \hat{\beta}_l l_{it} - \hat{\beta}_k k_{it} - \hat{\beta}_m m_{it}. \tag{6}$$

- Output is deflated by five-digit producer price indices.
- Domestic and imported intermediate inputs are deflated separately.
- The separate deflators matter especially during the Asian crisis, when imported input prices rose faster.

Interpretation. TFP is measured from revenue data, so the authors later spend substantial effort checking that their results are not just picking up markups or price mismeasurement.

2.7 Trade-status variables and firm controls

- **FM:** indicator that the firm imports any intermediate inputs.
- **Import share:** imported inputs divided by total inputs.
- **FX:** indicator that the firm exports any output.
- **Export share:** exports divided by output.
- **FF:** indicator that foreign ownership is at least 10%.
- **Exit:** indicator that the plant exits in the next period.

Interpretation. These variables are used both as direct controls and to test whether the tariff effects are really about importing, exporting, or foreign ownership.

2.8 Other key controls

- **Herfindahl index:** concentration at the four-digit industry level.
- **Highly concentrated industry:** Herfindahl above the 75th percentile (0.25).
- **Switch indicator:** whether a firm switches its main product category.
- **Asian crisis dummy:** years 1997 and 1998.
- **Trade-weighted exchange rate (TWI):** interacted with importer and exporter status.
- **Island-year fixed effects:** Sumatra, Java, Kalimantan, Sulawesi, and outer islands by year.

Interpretation. These controls are central to the paper's credibility, especially because revenue-based productivity can be contaminated by markups and crisis-driven price movements.

2.9 Descriptive facts

Table 3 reports the main descriptive moments:

- Mean output tariff: **0.159**; mean input tariff: **0.101**.
- Mean log TFP (Olley–Pakes): **1.639**.
- Mean import share: **0.098**; **20.7%** of observations are importers.
- Mean export share: **0.118**; **16.7%** of observations are exporters.
- Mean foreign share: **0.048**; **6.5%** of observations are foreign-owned by the paper's threshold.
- The crisis years represent about **19.8%** of observations.
- Mean Herfindahl index: **0.069**; only **5.3%** of observations lie in highly concentrated industries.
- About **6.3%** of firms exit in the following year.

Interpretation. Importing is not the norm in the sample, which is precisely why the importer interaction is so informative about direct imported-input gains.

3 Models and empirical specifications (all equations + interpretation)

3.1 Step 1: estimating productivity

The first-stage production function is estimated separately for each three-digit industry using the Olley–Pakes methodology. The reason for working at the three-digit rather than five-digit level is practical: the Olley–Pakes approach requires positive investment and enough within-industry observations to estimate a production function sensibly.

Interpretation. The paper accepts less industry detail in the first-stage productivity estimation in exchange for a more credible correction for simultaneity and exit.

3.2 Why Olley–Pakes instead of simple OLS

OLS would be biased because firms choose labor and materials after observing productivity shocks, and low-productivity firms are more likely to exit. Olley–Pakes addresses:

- simultaneity between productivity shocks and input choices;
- selection from firm exit.

The paper also modifies Olley–Pakes to let importing and exporting decisions act as additional state variables.

Interpretation. This is a major innovation of the paper: importing is not treated as exogenous when measuring productivity.

3.3 Modified Olley–Pakes structure

The investment rule is allowed to depend on capital, productivity, import status, and export status:

$$I_{it} = i_t(k_{it}, v_{it}, FM_{it}, FX_{it}). \tag{7}$$

Inverting this gives productivity as a function of observables:

$$v_{it} = h_t(k_{it}, I_{it}, FM_{it}, FX_{it}). \tag{8}$$

Substituting this into the production function yields a first-stage equation estimated nonparametrically.

Interpretation. The procedure is meant to purge TFP estimates of the fact that more productive firms are also more likely to import and export.

3.4 Step 2: baseline tariff regression

Using the plant-level TFP measures, the paper estimates:

$$\text{tfp}_{it}^{k} = \gamma_0 + a_i + a_{lt} + \gamma_1(\text{output tariff})_{kt} + \gamma_2(\text{input tariff})_{kt} + \gamma_3(\text{input tariff})_{kt} FM_{it} + \gamma_4 FM_{it} + e_{it}. \tag{9}$$

Here:

- a_i are firm fixed effects;
- a_{lt} are island-year fixed effects.

Interpretation.

- $\gamma_1 < 0$ captures the import-competition effect of lower output tariffs.
- $\gamma_2 < 0$ captures average gains from lower input tariffs, including possible spillovers to nonimporters.
- $\gamma_3 < 0$ captures extra gains for importing firms.

3.5 Expected signs and economic meaning

- A negative γ_1 means tougher import competition raises efficiency.
- A negative γ_2 means lower input tariffs help even nonimporters, perhaps through domestic spillovers or competition among input suppliers.

- A negative and significant γ_3 is the paper's most important coefficient because it says direct imported-input users gain the most.
- The coefficient on FM itself should be positive if importers are more productive on average.

3.6 Standard errors and panel structure

- The second-stage regression is estimated by OLS with a long firm panel.
- Standard errors are clustered at the firm level.
- The authors also report what happens under industry-year clustering; most qualitative conclusions survive.

Interpretation. The main evidence comes from within-firm changes over time as tariffs move differently across industries.

3.7 Alternative productivity specifications

The paper also estimates the tariff effects using:

- labor productivity: ln(real value added per worker);
- one-stage OLS with Cobb–Douglas production functions;
- two-stage OLS TFP estimates;
- translog production functions.

Interpretation. If the same tariff pattern appears regardless of how productivity is measured, it is less likely to be an artifact of one estimator.

3.8 Alternative panel transformations

The paper re-estimates the model using:

- levels with industry fixed effects;
- two-period differences;
- five-period differences;
- nine-period differences.

Interpretation. These are alternative ways to eliminate or reduce unobserved heterogeneity and serial dependence.

3.9 IV specification for tariff endogeneity

To address possible tariff endogeneity, the paper instruments five-period changes in tariffs using:

- 1991 tariff levels,
- an exclusion dummy for product codes exempted from the WTO tariff-reduction commitment,
- the proportion of low-skilled workers at the industry level,
- and interactions of initial tariffs with persistent importer indicators.

Interpretation. The IV design is not the core identification strategy, but it checks whether politically endogenous protection overturns the results.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline estimate

Identification comes from differential tariff changes across detailed industries over time, observed within the same plant. The firm fixed effects absorb time-invariant plant characteristics, and island-year fixed effects absorb broad regional shocks.

Interpretation. The key comparison is: when a plant's industry experiences larger input-tariff cuts than before, does that plant's measured productivity rise more than otherwise?

4.2 Why disaggregated tariffs are essential

The paper stresses that output and input tariffs need to be measured at a high level of disaggregation.

- At three-digit aggregation, output and input tariffs become highly collinear because industries often use inputs from within the same broad category as their own output.
- At five-digit aggregation, the estimates are much more stable.

Interpretation. If tariffs are too aggregated, one cannot cleanly separate competition effects from imported-input effects.

4.3 Why reverse causality is limited

- The main tariff cuts were tied to Indonesia's WTO commitments beginning in 1995.
- Mobarak and Purbasari find that political connections in Indonesia did not affect tariff rates.
- Firm fixed effects already remove time-invariant industry lobbying strength.

Interpretation. The case for severe tariff endogeneity is weaker here than in many political-economy settings.

4.4 How the paper addresses markup concerns

Revenue-based TFP may partly reflect markups rather than physical productivity. The paper therefore:

- includes a Herfindahl index;
- interacts tariff variables with a high-concentration dummy.

The result is revealing:

- lower output tariffs raise productivity only in competitive industries;
- lower input tariffs raise importer productivity in both concentrated and competitive industries.

Interpretation. The output-tariff channel is more plausibly tied to import-competition pressure on markups, while the input-tariff channel looks broader and harder to dismiss as a pure markup story.

4.5 How the paper addresses product-switching concerns

If firms switch from low-productivity to high-productivity products after liberalization, measured TFP could rise even without within-product efficiency gains. The paper adds a switching dummy and interactions with tariffs.

- Product switching is positively associated with measured productivity.
- Some of the output-tariff effect is related to switching.
- The input-tariff effect for importers is not.

Interpretation. The direct imported-input channel does not appear to be driven by firms merely changing product categories.

4.6 Foreign firms, exporters, and import licenses

The paper checks whether the results are really about foreign ownership, exporting, or special import rights:

- excluding foreign firms leaves the main result intact;
- allowing special effects for importer-exporters adds little;
- excluding firms with import licenses leaves the main result essentially unchanged.

Interpretation. The productivity gain from lower input tariffs is not simply a foreign-firm or exporter phenomenon.

4.7 Asian crisis and exchange-rate confounds

The Asian financial crisis is a major threat because large depreciations could distort revenue-based productivity. The paper therefore:

- re-estimates the model only through 1996;
- adds exchange rates interacted with importers and exporters;
- interacts tariff terms with a crisis dummy.

Interpretation. These exercises reduce the magnitude somewhat, but the importer-specific input-tariff effect remains positive and sizable.

4.8 What the paper can and cannot distinguish

- The paper *can* distinguish output-tariff from input-tariff effects, and importer from nonimporter effects.
- It *cannot* separately identify whether the imported-input gains come from higher quality, more varieties, embodied foreign technology, or learning.
- It also cannot fully separate physical productivity from markups using only revenue data.

Interpretation. The paper is strongest on *who gains and from which tariff margin*, but weaker on the exact micro-channel of imported-input gains.

4.9 Relation to the literature

Earlier empirical work mainly studied output tariffs and import competition. This paper's distinctive contribution is to show that:

- input tariffs matter separately,
- their effects are larger,
- and the largest gains accrue to firms that actually import inputs.

5 Tables and figures

5.1 Table 1 and Figure 2: the tariff environment

Read:

- Average output tariffs fell from about **20.88%** in 1991 to **8.44%** in 2001.
- Average input tariffs fell from about **13.71%** to **5.94%**.

Table 1—Tariffs in Indonesia, 1991–2001

Industry	Tariff type	1991	1993	1995	1997	1999	2001
31 food	output	20.84	20.50	20.62	19.19	15.95	15.94
	input	13.86	13.60	9.83	8.68	7.12	6.95
32 textile clothing	output	27.35	26.65	20.19	16.53	12.68	9.43
	input	17.59	17.38	13.25	10.76	8.87	6.27
33 wood	output	24.20	24.10	17.95	12.32	9.43	6.91
	input	10.24	10.09	6.52	4.32	3.57	2.90
34 paper	output	21.21	19.76	10.09	7.04	4.31	4.03
	input	17.56	16.30	9.42	6.86	4.81	4.18
35 chemicals	output	15.60	14.93	12.05	10.11	8.31	6.92
	input	11.14	11.05	9.00	7.57	6.26	5.16
36 metal	output	23.04	21.84	10.62	7.46	6.40	5.65
	input	14.81	13.94	9.52	7.95	6.61	5.64
37 machinery	output	11.50	9.72	8.08	7.32	6.85	5.77
	input	9.80	9.94	7.82	7.32	6.88	6.15
38 electrical	output	18.89	18.56	14.69	11.01	7.75	6.69
	input	13.84	13.53	10.25	8.32	7.26	6.26
39 other	output	32.50	31.57	22.11	17.70	14.28	10.98
	input	15.94	15.37	11.25	9.17	7.67	6.17
All	output	20.88	20.29	15.58	12.51	9.76	8.44
	input	13.71	13.40	9.92	8.24	6.91	5.94

Table 2—Coefficients of the Production Function

Industry	Labor		Materials		Capital	
	OLS	OP	OLS	OP	OLS	OP
Food products (311)	0.304	0.273	0.747	0.708	0.058	0.067
Food products, nes[a] (312)	0.421	0.335	0.494	0.467	0.172	0.132
Beverages (313)	0.965	0.818	0.353	0.346	0.166	0.175
Tobacco (314)	0.159	0.105	0.875	0.875	0.036	0.000
Textiles (321)	0.249	0.212	0.728	0.708	0.058	0.064
Clothing (322)	0.277	0.253	0.743	0.724	0.039	0.070
Leather goods, nes[a] (323)	0.334	0.321	0.718	0.702	0.026	0.003
Leather footwear (324)	0.392	0.351	0.643	0.619	0.017	0.002
Wood and cork, except furniture (331)	0.296	0.276	0.698	0.677	0.046	0.061
Furniture (332)	0.303	0.285	0.690	0.677	0.052	0.046
Paper and paper products (341)	0.281	0.230	0.739	0.730	0.044	0.018
Printing, publishing, and allied industries (342)	0.419	0.292	0.645	0.657	0.053	0.063
Industrial chemicals (351)	0.312	0.173	0.561	0.497	0.150	0.178
Other chemical products (352)	0.409	0.376	0.641	0.607	0.094	0.121
Rubber products (355)	0.221	0.223	0.717	0.694	0.049	0.045
Plastic products, nes[a] (356)	0.247	0.203	0.745	0.717	0.049	0.056
Pottery, china, and earthware (361)	0.353	0.377	0.583	0.498	0.145	0.196
Glass and glass products (362)	0.381	0.278	0.668	0.640	0.059	0.120
Cement (363)	0.358	0.251	0.713	0.706	0.062	0.128
Clay products (364)	0.544	0.517	0.422	0.367	0.137	0.115
Other nonmetallic mineral products (369)	0.448	0.364	0.578	0.518	0.164	0.222
Iron and steel industries (371)	0.259	0.248	0.787	0.755	0.015	0.045
Nonferrous metal basic industries (372)	0.364	0.182	0.691	0.664	0.124	0.174
Fabricated metal products, except machinery (381)	0.315	0.285	0.714	0.701	0.040	0.031
Nonelectrical machinery (382)	0.327	0.268	0.693	0.677	0.080	0.044
Electrical machinery (383)	0.289	0.293	0.737	0.713	0.044	0.096
Transport equipment (384)	0.384	0.312	0.671	0.639	0.051	0.143
Professional, scientific, and equipment (385)	0.384	0.312	0.671	0.639	0.051	0.143
Miscellaneous manufacturing (390)	0.390	0.346	0.620	0.589	0.074	0.133

[a] "nes" refers to "not elsewhere classified."

- Figure 2 shows that industries with the highest initial tariffs generally experienced the largest cuts.
- Output tariffs were usually higher than input tariffs, consistent with tariff escalation.

Economic message. Indonesia provides sharp and economically meaningful tariff variation, especially after 1995, and the variation is richer than a simple common trend.

5.2 Figure 1 and Table 2: price deflators and production coefficients

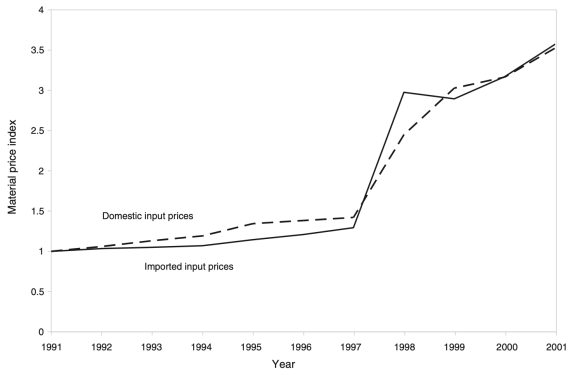

Figure 1. Input Price Indices

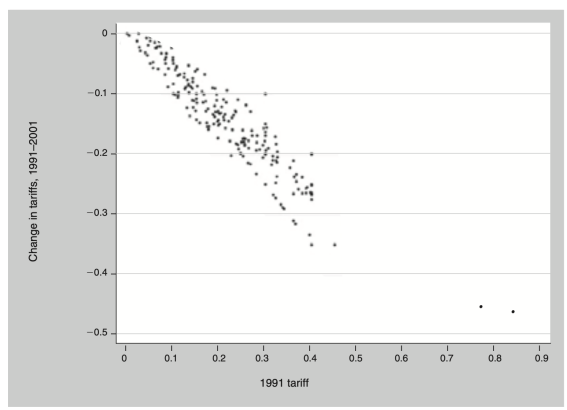

Figure 2. Change in Tariffs, 1991–2001, Relative to Initial Levels

Note: Industries that experienced an increase in their tariff over the sample period are excluded from the figure. These are industries 31161, 31169, 31310, and 31320.

Read:

- Figure 1 shows that domestic and imported input price indices mostly move together, but imported input prices jump more sharply during the Asian crisis.
- Table 2 shows that OLS generally overstates labor and materials coefficients relative to Olley–Pakes, consistent with simultaneity bias.

Economic message. Separate input deflators and a corrected production-function estimator are both important before one even gets to the tariff regressions.

5.3 Table 3: summary statistics

Table 3—Summary Statistics

Variable	Observations	Mean	Standard deviation
Output tariff	170,741	0.159	0.113
Output $\text{tariff}_t - \text{Output tariff}_{t-2}$	111,107	−0.037	0.057
Output $\text{tariff}_t - \text{Output tariff}_{t-5}$	56,320	−0.106	0.092
Output $\text{tariff}_t - \text{Output tariff}_{t-9}$	6,089	−0.163	0.116
Output tariff–3 digit	170,741	0.166	0.138
Output tariff on the basis of the IO table	170,741	0.110	0.207
Output weighted tariff	170,741	0.163	0.297
Input tariff	170,741	0.101	0.062
Input $\text{tariff}_t - \text{input tariff}_{t-2}$	111,107	−0.018	0.031
Input $\text{tariff}_t - \text{input tariff}_{t-5}$	56,320	−0.053	0.050
Input $\text{tariff}_t - \text{input tariff}_{t-9}$	6,089	−0.082	0.064
Input tariff–3 digit	170,741	0.129	0.061
Input tariff on the basis of the IO table–1995	170,741	0.078	0.095
Input tariff on the basis of the IO table–1998	170,741	0.068	0.078
Input weighted tariff	170,741	0.099	0.227
Log real output	170,468	8.160	2.106
ln(TFP)–Olley-Pakes	170,741	1.639	0.671
$\ln(\text{TFP}_t)$–$\ln(\text{TFP}_{t-5})$–Olley-Pakes	56,320	0.124	0.527
ln(TFP)–OLS	170,741	1.331	0.597
ln(TFP)–no foreign	170,741	1.653	0.665
ln(Value added per worker)	165,025	3.199	1.293
ln(L)	170,740	4.247	1.230
ln(K)	169,527	7.069	2.275
ln(K/L)	170,740	2.816	1.721
ln(Materials)	170,570	7.222	2.307
ln(Total inputs)	170,570	12.969	2.377
Import share	170,741	0.098	0.250
FM = 1 if import share ≥ 0	170,741	0.207	0.405
$\text{FM}_t - \text{FM}_{t-2}$	111,107	−0.009	0.263
$\text{FM}_t - \text{FM}_{t-5}$	56,320	−0.019	0.341
$\text{FM}_t - \text{FM}_{t-9}$	6,089	−0.049	0.408
High FM[a]	170,741	0.102	0.303
Export share	170,741	0.118	0.297
FX = 1 if export share > 0	170,741	0.167	0.373
$\text{FX}_t - \text{FX}_{t-2}$	111,107	−0.010	0.354
$\text{FX}_t - \text{FX}_{t-5}$	56,320	−0.046	0.393
$\text{FX}_t - \text{FX}_{t-9}$	6,089	0.002	0.394
Foreign share	170,741	0.048	0.190
FF = 1 if foreign share ≥ 0.1	170,741	0.065	0.247
$\text{FF}_t - \text{FF}_{t-2}$	111,107	0.001	0.125
$\text{FF}_t - \text{FF}_{t-5}$	56,320	0.002	0.153
$\text{FF}_t - \text{FF}_{t-9}$	6,089	0.005	0.167
ln(TWI)	170,741	−0.737	0.601
$\ln(\text{TWI}_t) - \ln(\text{TWI}_{t-5})$	56,320	−0.999	0.384
Switch = 1 if firm switches products	170,741	0.148	0.355
Crisis dummy = 1 if year = 1997 or 1998	170,741	0.198	0.399
Herfindahl index–4-digit level	170,741	0.069	0.091
$\text{Herf}_t - \text{Herf}_{t-2}$	111,107	−0.002	0.067
$\text{Herf}_t - \text{Herf}_{t-5}$	56,320	0.002	0.082
$\text{Herf}_t - \text{Herf}_{t-9}$	6,089	−0.016	0.117
Highly concentrated industry (Herfindahl > 0.25)	170,741	0.053	0.223
Exit = 1 if firm exits next year	170,741	0.063	0.243

[a] High FM indicates importing firms in industries with more than 40 percent of firms.

Read:

- There are **170,741** plant-year observations.
- Mean import share is **9.8%**; mean export share is **11.8%**.
- About **20.7%** of observations involve importing firms and **16.7%** involve exporters.
- Mean output tariff is **15.9%**; mean input tariff is **10.1%**.

Economic message. The sample has enough importer/nonimporter variation and tariff variation to make the interaction design meaningful.

5.4 Table 4: baseline productivity results

Read:

- Column 1, using only output tariffs, gives a coefficient of about −0.206. A 10 percentage point fall in output tariffs is associated with a roughly **2.1%** increase in productivity.
- Column 2 adds input tariffs. The output-tariff coefficient falls to about −0.070, while the input-tariff coefficient is about −0.441.
- Column 3 adds the interaction with importer status. The interaction term is about −0.914.
- Combining the coefficients implies that a 10 percentage point fall in input tariffs raises productivity by about **12%** for importing firms and about **3%** for nonimporters.
- Importers are also more productive on average: the coefficient on FM is about **0.092**.

Economic message. This is the paper's main result: ignoring input tariffs overstates the importance of output-tariff competition effects, while the largest gains come from cheaper and more accessible inputs.

Table 4—Basic Results

Dependent variable: $\ln(\text{TFP}_{it})$	(1)	(2)	(3)	(4)	(5)	(6)
Output tariff_t^k	−0.206***	−0.070*	−0.092**	−0.096**	−0.096**	−0.095**
	(0.033)	(0.042)	(0.043)	(0.043)	(0.043)	(0.043)
Input tariff_t^k		−0.441***	−0.318***	−0.315***	−0.315***	−0.325***
		(0.062)	(0.063)	(0.063)	(0.063)	(0.063)
Input $\text{tariff}_t^k \times \text{FM}_{it}$			−0.914***	−0.899***	−0.896***	
			(0.086)	(0.086)	(0.086)	
$\text{FM}_{it} = 1$ if import share > 0			0.092***	0.091***	0.089***	
			(0.012)	(0.012)	(0.012)	
Input $\text{tariff}_t^k \times$ import share_{it}						−1.908***
						(0.164)
Import share_{it}						0.233***
						(0.024)
$\text{FX}_{it} = 1$ if export share > 0					−0.010**	
					(0.005)	
Export share_{it}						−0.008
						(0.006)
$\text{FF}_{it} = 1$ if foreign share $\geqq 0.1$					0.070***	
					(0.017)	
$\text{Foreign share}_{it}$						0.079***
						(0.023)
$\text{Exit}_{it} = 1$ if firm exits in $t+1$				−0.040***	−0.040***	−0.040***
				(0.004)	(0.004)	(0.004)
Island × year effects	yes	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes	yes
Observations	170,741	170,741	170,741	170,741	170,741	170,741
R-squared	0.80	0.80	0.80	0.80	0.80	0.80

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except *output tariff* in columns 2 through 6 becomes insignificant.
*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

Table 5—Alternative Productivity Measures

Dependent variable:	$\ln(\text{real value added per worker}_{it})$			$\ln(\text{real output}_{it})$ Cobb-Douglas technology		$\ln(\text{TFP}_{it})$ Cobb-Douglas technology		$\ln(\text{TFP}_{it})$ Translog technology	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Output tariff_t^k	−0.553***	−0.533***	−0.267***	−0.212***	−0.158***	−0.162***	−0.058	−0.236***	−0.167***
	(0.065)	(0.065)	(0.073)	(0.032)	(0.043)	(0.030)	(0.040)	(0.031)	(0.040)
Input tariff_t^k			−0.793***		−0.168***		−0.302***		−0.186***
			(0.124)		(0.063)		(0.058)		(0.059)
Input $\text{tariff}_t^k \times \text{FM}_{it}$			−1.186***		−0.775***		−0.743***		−0.787***
			(0.160)		(0.088)		(0.082)		(0.081)
$\text{FM}_{it} = 1$ if import share > 0			0.233***		0.090***		0.066***		0.090***
			(0.022)		(0.012)		(0.011)		(0.011)
$\text{FX}_{it} = 1$ if export share > 0			0.007		−0.003		−0.007		0.003
			(0.009)		(0.005)		(0.005)		(0.005)
$\text{FF}_{it} = 1$ if foreign share $\geqq 0.1$			0.179***		0.072***		0.053***		0.081***
			(0.030)		(0.017)		(0.017)		(0.016)
$\text{Exit}_{it} = 1$ if firm exits in $t+1$			−0.083***		−0.032***		−0.031***		−0.034***
			(0.009)		(0.005)		(0.004)		(0.004)
$\ln(\text{K/L})_{it}$		0.058***	0.060***						
		(0.003)	(0.003)						
$\ln(\text{real inputs}_{it})$ × prod5d [a]				yes	yes				
Island × year effects	yes	yes	yes	yes	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	165,025	165,025	165,025	170,741	170,741	170,741	170,741	169,217	169,217
R-squared	0.81	0.81	0.81	0.74	0.75	0.80	0.80	0.98	0.98

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except *output tariff* in column 5 becomes insignificant, *FX* in column 9 becomes insignificant, and *input tariff* in column 9 becomes insignificant, but the interactive *input tariff* × *FM* remains significant in all columns.
[a] The inputs, labor, real capital, and real materials, are all interacted with five-digit industry dummies.
*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

5.5 Table 5: alternative productivity measures

Read:

- Using labor productivity, the output-tariff coefficient is large when entered alone (around −0.533) but falls sharply once input tariffs are included (to about −0.267).
- The importer interaction with input tariffs remains strongly negative, around −1.186 in the labor-productivity specification.
- The same qualitative pattern appears in one-stage Cobb–Douglas and translog TFP specifications.

Economic message. The central pattern is not a quirk of the specific TFP estimator. It survives across very different productivity measures.

5.6 Table 6: channels and alternative explanations

Read:

- The Herfindahl index is negative: more concentrated industries have lower measured productivity on average.
- The interaction of output tariffs with high concentration is positive and significant, implying output-tariff gains mainly accrue in competitive industries.

Table 6—Channels

Dependent variable: $\ln(\text{TFP}_{ijt})$	Mark-ups		Product switchers		No foreign firms	Exporters	No licensed firms
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Output tariff^k_t	−0.094**	−0.138***	−0.097**	−0.090***	−0.048	−0.095**	−0.088**
	(0.043)	(0.039)	(0.043)	(0.028)	(0.043)	(0.043)	(0.043)
Output tariff^k_t × high concentration^k_t		0.231***					
		(0.049)					
Output tariff^k_t × switch$_{ijt}$				−0.069*			
				(0.039)			
Input tariff^k_t	−0.318***	−0.286***	−0.310***	−0.315***	−0.359***	−0.318***	−0.318***
	(0.063)	(0.061)	(0.063)	(0.042)	(0.063)	(0.063)	(0.063)
Input tariff^k_t × FM$_{ijt}$	−0.896***	−0.908***	−0.887***	−0.882***	−0.655***	−0.876***	−0.892***
	(0.086)	(0.086)	(0.086)	(0.064)	(0.089)	(0.090)	(0.087)
Input tariff^k_t × FM$_{ijt}$ × high concentration^k_t		0.068					
		(0.134)					
Input tariff^k_t × FM$_{ijt}$ × switch$_{ijt}$				−0.003			
				(0.085)			
Input tariff^k_t × FM$_{ijt}$ × FX$_{ijt}$						−0.048	
						(0.072)	
FM$_{ijt}$ = 1 if import share > 0	0.090***	0.090***	0.089***	0.088***	0.058***	0.090***	0.092***
	(0.012)	(0.012)	(0.012)	(0.009)	(0.012)	(0.012)	(0.012)
FX$_{ijt}$ = 1 if export share > 0	−0.010**	−0.010**	−0.010**	−0.010**	−0.007	−0.007	−0.007
	(0.005)	(0.005)	(0.005)	(0.004)	(0.005)	(0.005)	(0.005)
FF$_{ijt}$ = 1 if foreign share > 0.1	0.070***	0.070***	0.070***	0.070***			
	(0.017)	(0.017)	(0.017)	(0.014)			
Exit$_{ijt}$ = 1 if firm exits in $t+1$	−0.040***	−0.040***	−0.040***	−0.040***	−0.047***	−0.040***	−0.040***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Herfindahl index^k_t	−0.080***	−0.185***	−0.081***	−0.080***	−0.073***	−0.080***	−0.081***
	(0.021)	(0.030)	(0.021)	(0.019)	(0.021)	(0.021)	(0.020)
Switch$_{ijt}$			0.029***	0.038***			
			(0.006)	(0.008)			
Island × year effects	yes	yes	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes
Observations	170,741	170,741	170,741	170,741	159,640	170,741	167,953
R-squared	0.80	0.80	0.80	0.80	0.80	0.80	0.80

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except *output tariff* in columns 1, 3, 6, and 7 becomes insignificant, *output tariff* in column 2 is significant only at the 10 percent level, *Herfindahl index* in columns 1, 5, and 6 becomes insignificant, and *Herfindahl index* in columns 3 and 7 is significant only at the 10 percent level.

*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

- The interaction of input tariffs, importer status, and high concentration is insignificant.
- The product-switching interaction explains part of the output-tariff effect but not the input-tariff importer effect.
- Excluding foreign firms still leaves a large importer interaction (about −0.655).

Economic message. The imported-input channel is robust to markup concerns, product switching, foreign ownership, exporting status, and import licensing.

5.7 Table 7: alternative tariff constructions

Read:

- At the three-digit level, coefficients become unstable because output and input tariffs are too collinear.
- Dropping the liquor industry changes three-digit results sharply, illustrating the fragility of over-aggregated tariffs.
- The five-digit results are stable when the liquor industry is excluded.
- Import-weighted tariffs produce similar patterns, though some output-tariff coefficients weaken.
- Input tariffs built from 1995 cost shares look very similar to those built from 1998 cost shares.

Economic message. High tariff disaggregation is essential, and the baseline results are not an artifact of using 1998 crisis-year input shares.

Table 7—Alternative Tariffs

Dependent variable: $\ln(\text{TFP}_{ijt})$	Three-digit tariffs		Three-digit tariffs –no liquor		Five-digit tariffs –no liquor		Five-digit tariffs –import weighted		I/O code tariffs with 1995 cost shares	I/O code tariffs with 1998 cost shares
	(1)	(2)	(3)	(4)	(5)	(6)	(7)		(8)	(9)
Output tariff^k_t	−0.059	−0.007	−0.535***	−0.448***	−0.128***	−0.044***	0.003		0.002	0.009
	(0.038)	(0.044)	(0.035)	(0.066)	(0.042)	(0.003)	(0.018)		(0.023)	(0.027)
Input tariff^k_t		−0.203*		0.399***	−0.428***		−0.462***		−0.507***	−0.539***
		(0.104)		(0.110)	(0.065)		(0.023)		(0.048)	(0.067)
Input tariff^k_t × FM$_{ijt}$							−0.946***		−0.329***	−0.420***
							(0.075)		(0.050)	(0.059)
FM$_{ijt}$ = 1 if import share > 0		−0.007		−0.005		−0.006		0.083***	0.023***	0.024***
		(0.007)		(0.007)		(0.007)		(0.010)	(0.008)	(0.009)
FX$_{ijt}$ = 1 if export share > 0		−0.013***		−0.013***		−0.013***		−0.01**	−0.010**	−0.010**
		(0.005)		(0.005)		(0.005)		(0.005)	(0.005)	(0.005)
FF$_{ijt}$ = 1 if foreign share ≥ 0.1		0.070***		0.070***		0.074***		0.069***	0.070***	0.071***
		(0.017)		(0.017)		(0.017)		(0.017)	(0.017)	(0.017)
Herfindahl index^k_t		−0.075***		−0.073***		−0.078***		−0.083***	−0.072***	−0.061***
		(0.021)		(0.021)		(0.021)		(0.021)	(0.021)	(0.021)
Exit$_{ijt}$ = 1 if firm exits in $t+1$		−0.042***		−0.042***		−0.042***		−0.040***	−0.041***	−0.042***
		(0.004)		(0.004)		(0.004)		(0.004)	(0.004)	(0.004)
Island × year effects	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	170,741	170,741	168,640	168,640	168,840	168,840	170,741	170,741	170,741	170,741
R-squared	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except that *output tariff* in column 5 becomes insignificant, *input tariff* in columns 2 and 7 becomes insignificant, *Herfindahl index* in columns 2–5 and 8–9 becomes insignificant, and *Herfindahl index* in column 7 is significant only at the 10 percent level.

*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

Table 8—Asian Financial Crisis

Dependent variable: $\ln(\text{TFP}_{ijt})$	1995–1996 Pre–Asian crisis		1990–2001 Full sample			
	(1)	(2)	(3)	(4)	(5)	(6)[a]
Output tariff^k_t	−0.015	−0.018	−0.103**	−0.107**	−0.114***	−0.120***
	(0.044)	(0.044)	(0.042)	(0.043)	(0.043)	(0.043)
Input tariff^k_t	−0.363***	−0.369***	−0.334***	−0.354***	−0.328***	−0.326***
	(0.119)	(0.118)	(0.062)	(0.062)	(0.062)	(0.062)
Input tariff^k_t × FM$_{it}$	−0.118	−0.057	−0.875***	−0.262**	−0.261**	−0.477**
	(0.109)	(0.109)	(0.086)	(0.108)	(0.108)	(0.225)
Input tariff^k_t × FM$_{it}$ × high FM^k		−0.693***				
		(0.130)				
FM$_{it}$ = 1 if import share > 0	−0.010	0.012	0.085***	−0.024	−0.024	−0.023
	(0.017)	(0.018)	(0.012)	(0.018)	(0.018)	(0.038)
FX$_{it}$ = 1 if export share > 0	0.023***	0.023***	−0.010**	−0.026***	−0.026***	−0.035***
	(0.007)	(0.007)	(0.005)	(0.007)	(0.007)	(0.009)
FF$_{it}$ = 1 if foreign share ≥ 0.1	0.061***	0.062***	0.070***	0.068***	0.068***	0.071***
	(0.021)	(0.023)	(0.017)	(0.017)	(0.017)	(0.023)
ln(TW1^k_t) × FM$_{it}$				−0.064***	−0.063***	−0.134***
				(0.009)	(0.009)	(0.017)
ln(TW1^k_t) × FX$_{it}$				−0.025***	−0.025***	−0.039***
				(0.007)	(0.007)	(0.008)
Output tariff^k_t × crisis dummy$_t$				0.031	0.028	0.030
				(0.040)	(0.040)	(0.040)
Input tariff^k_t × crisis dummy$_t$				0.009	0.043	0.026
				(0.079)	(0.079)	(0.079)
Input tariff^k_t × FM$_{it}$ × crisis dummy$_t$				0.738***	0.048	0.146
				(0.067)	(0.069)	(0.112)
Herfindahl index^k_t	−0.011	−0.019	−0.080***	−0.078***	−0.076***	−0.081***
	(0.024)	(0.024)	(0.021)	(0.021)	(0.021)	(0.021)
Exit$_{it}$ = 1 if firm exits in $t+1$	−0.021***	−0.021***	−0.040***	−0.038***	−0.038***	−0.039***
	(0.006)	(0.006)	(0.004)	(0.004)	(0.004)	(0.004)
Year fixed effects	yes	yes	yes	yes	yes	yes
Observations	88,442	88,442	170,741	170,741	170,741	170,741
R-squared	0.87	0.87	0.80	0.80	0.80	0.80

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except that *output tariff* in columns 3–6 becomes insignificant, *input tariff* in columns 1 and 2 becomes insignificant, *input tariff* × *FM* × *crisis* in column 3 becomes insignificant, *Herfindahl index* in columns 3–5 becomes insignificant, and *Herfindahl index* in column 6 is significant only at the 10 percent level.

[a] In column 6, *FM* is import share, *FX* is export share, and *FF* is foreign share.

*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

5.8 Table 8: Asian crisis

Read:

- In the full sample with exchange-rate interactions, the importer-specific input-tariff coefficient falls from the baseline magnitude to roughly -0.262.
- That still implies about a **6%** productivity gain for importers from a 10 percentage point fall in input tariffs after accounting for exchange-rate effects.
- In the pre-1997 sample, the estimates are weaker, largely because most tariff liberalization occurred after 1995 and there is much less variation.

Economic message. The Asian crisis explains part of the baseline magnitude but does not overturn the central conclusion that input-tariff cuts help importers substantially.

5.9 Table 9: alternative panel estimators

Read:

- In two-period differences, the importer interaction remains negative and significant (around -0.515).
- In five-period differences, the importer interaction is still around -0.455; the output-tariff coefficient is smaller and sometimes insignificant.
- In nine-period differences, the importer interaction is very large, around -0.902, though the sample is much smaller.

Economic message. The finding that importers gain most from lower input tariffs is not tied to one specific panel estimator.

5.10 Table 10: tariff endogeneity and IV

Read:

- IV estimates on five-period differences make the output-tariff coefficient much larger in absolute value (roughly -0.63 to -0.75).
- The average input-tariff coefficient for nonimporters often becomes insignificant.
- The interaction of input tariffs with importer status remains strongly negative, around -1.27 to -1.32.
- The Hansen J statistics have comfortable p-values from about **0.14** to **0.72**.

Economic message. If anything, OLS may understate the direct gains to importers and overstate gains to nonimporters.

Table 9—Alternative Econometric Specifications

Dependent variable: $\ln(\text{TFP}_{it})$	Levels		Two-period difference		Five-period difference		Nine-period difference	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔOutput tariff^k_t	−0.224***	−0.127***	−0.140***	−0.014	−0.204***	−0.085*	−0.386***	−0.302***
	(0.032)	(0.041)	(0.027)	(0.036)	(0.040)	(0.050)	(0.072)	(0.096)
ΔInput tariff^k_t		−0.279***		−0.336***		−0.365***		−0.158
		(0.059)		(0.054)		(0.071)		(0.145)
ΔInput tariff$^k_t \times \text{FM}_{i,t-j}$		−0.489***		−0.515***		−0.455***		−0.902**
		(0.100)		(0.139)		(0.174)		(0.411)
ΔFM_{it}	0.019**	0.092***	0.000	0.003	0.028***	0.029***	0.038	0.038
	(0.007)	(0.017)	(0.007)	(0.007)	(0.010)	(0.010)	(0.024)	(0.024)
ΔFX_{it}	0.066***	0.068***	0.012**	0.011**	0.026***	0.023***	0.026	0.024
	(0.007)	(0.007)	(0.005)	(0.005)	(0.009)	(0.009)	(0.024)	(0.024)
ΔFF_{it}	0.247***	0.248***	0.079***	0.080***	0.068***	0.069***	0.117**	0.118**
	(0.011)	(0.011)	(0.016)	(0.016)	(0.024)	(0.024)	(0.057)	(0.057)
$\Delta\ln(\text{TWI}^k_t) \times \text{FM}_{i,t-j}$	−0.096***	−0.068***	−0.067***	−0.049***	−0.080***	−0.055***	−0.065***	−0.009
	(0.007)	(0.009)	(0.008)	(0.010)	(0.008)	(0.012)	(0.014)	(0.029)
$\Delta\ln(\text{TWI}^k_t) \times \text{FX}_{i,t-j}$	−0.004	−0.002	−0.075***	−0.072***	−0.064***	−0.060***	−0.037**	−0.035**
	(0.007)	(0.007)	(0.009)	(0.009)	(0.009)	(0.009)	(0.018)	(0.018)
ΔHerfindahl index^k_t	−0.132***	−0.139***	−0.002	0.000	−0.195***	−0.198***	0.071	0.048
	(0.023)	(0.023)	(0.019)	(0.019)	(0.030)	(0.030)	(0.063)	(0.066)
Exit_{it} = 1 if firm exits in $t+1$	−0.081***	−0.081***						
	(0.005)	(0.005)						
Island × year effects	yes	yes	yes	yes	yes	yes	island effects	
Industry fixed effects	yes	yes	no	no	no	no	no	no
Firm fixed effects	no	no	yes	yes	yes	yes	no	no
Observations	170,741	170,741	111,107	111,107	56,320	56,320	6,089	6,089
R-squared	0.55	0.55	0.01	0.01	0.02	0.02	0.01	0.02

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry-year level, all significant variables remain significant with p-values < 0.05, except *output tariff* in columns 2 and 6 becomes insignificant and *input tariff* in column 6 is significant only at the 10 percent level.
*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

Table 10—Endogeneity

Dependent variable: $\ln(\text{TFP}_{it})-\ln(\text{TFP}_{i,t-5})$ All variables in 5 period difference	(1)	(2)	(3)	(4)	(5)	(6)
ΔOutput tariff^k_t	−0.754***	−0.657***	−0.644***	−0.629***	−0.638***	−0.668***
	(0.067)	(0.071)	(0.071)	(0.071)	(0.072)	(0.082)
ΔInput tariff^k_t		−0.326***	−0.099	−0.106	−0.171	−0.159
		(0.106)	(0.123)	(0.123)	(0.115)	(0.131)
ΔInput tariff^k_t × $\text{FM}_{i,t-5}$			−1.274***	−1.324***		
			(0.415)	(0.413)		
ΔInput tariff^k_t × import $\text{share}_{i,t-5}$					−1.942***	−2.211***
					(0.724)	(0.814)
ΔFM_{it}			0.040***	0.038***		
			(0.010)	(0.010)		
ΔImport share_{it}					0.118***	0.097***
					(0.021)	(0.025)
ΔFX_{it}				0.020**		
				(0.009)		
ΔExport share_{it}					0.032***	0.033**
					(0.011)	(0.014)
ΔFF_{it}				0.070***		
				(0.024)		
ΔForeign share_{it}					0.001***	0.001
					(0.000)	(0.000)
$\Delta\ln(\text{TWI}^k_t) \times \text{FM}_{i,t-5}$	−0.078***	−0.076***	−0.025	−0.021		
	(0.007)	(0.007)	(0.022)	(0.022)		
$\Delta\ln(\text{TWI}^k_t)$ × import $\text{share}_{i,t-5}$					−0.078**	−0.029
					(0.039)	(0.044)
$\Delta\ln(\text{TWI}^k_t) \times \text{FX}_{i,t-5}$	−0.046***	−0.044***	−0.042***	−0.053***		
	(0.007)	(0.007)	(0.007)	(0.009)		
$\Delta\ln(\text{TWI}^k_t)$ × export $\text{share}_{i,t-5}$					−0.069***	−0.070***
					(0.011)	(0.014)
ΔHerfindahl index^k_t				−0.209***	−0.210***	−0.185***
				(0.030)	(0.030)	(0.034)
Island x year effects	yes	yes	yes	yes	yes	yes
Firm fixed effects	yes	yes	yes	yes	yes	yes
Weak instruments	F(3)=5712.42	$\chi^2(4)$=6528.68	$\chi^2(6)$=2081.14	$\chi^2(6)$=2082.70	$\chi^2(6)$=2196.36	$\chi^2(6)$=1609.52
Overidentification	2.80	2.08	2.80	2.78	1.36	5.48
Hansen J statistic	0.25	0.35	0.43	0.43	0.72	0.14
Observations	56,320	56,320	56,320	56,320	56,320	41,676

Notes: Robust standard errors corrected for clustering at the firm level in parentheses. If, instead, error terms were corrected for clustering at the industry/year level, all significant variables remain significant with p-values <0.05, except input tariff in column 2 becomes insignificant and *TWI × import share* in column 5 is significant only at the 10 percent level. Instruments in column (1) *output tariff*$_{1991}$, *exclusion dummy* = 1 if product excluded from commitment to reduce bound tariffs to 40 percent, *proportion of low-skilled workers*^k_t; (2) as in column 1, plus *input tariff*$^k_{1991}$; (3)–(5) as in column 2, plus *input tariff*$^k_{1991}$ if firm imported every year in the sample and a dummy if firm imported every year in the sample; (6) as in column 2, plus *input tariff*$^k_{1991}$ if the firm imported in 1991 and a dummy if the firm imported in 1991.
*** Significant at, or below, 1 percent.
** Significant at, or below, 5 percent.
* Significant at, or below, 10 percent.

6 Short summary

- The paper asks whether tariff cuts on inputs or outputs matter more for plant productivity.
- It uses Indonesian manufacturing census data and very detailed input-use information to construct industry-specific input tariffs.
- Productivity is estimated with a modified Olley–Pakes method that treats importing and exporting as endogenous state variables.
- Once input tariffs are included, the effect of output tariffs becomes much smaller.
- The large productivity gains come from lower input tariffs, especially for firms that directly import their inputs.
- The main result survives many robustness checks, including alternative productivity measures, tariff constructions, crisis controls, panel estimators, and IV.
- The paper cannot say exactly whether the imported-input channel works through quality, variety, embodied foreign technology, or learning, but it shows clearly that the imported-input channel is economically first order.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that lowering *input* tariffs has a large positive effect on firm productivity in Indonesian manufacturing, and that the effect is especially strong for firms that actually import intermediate inputs. Lower *output* tariffs also help, but much less once the input side is accounted for.

7.2 Economic intuition

Tariff cuts on outputs mainly make firms compete harder. Tariff cuts on inputs change what firms can actually use in production. When access to imported intermediates improves, firms can buy cheaper or better inputs, tap into foreign technology, and possibly use a broader range of varieties. Those direct imported-input gains dominate the standard competition effect.

7.3 Takeaway

For trade reform in developing countries, the most important productivity margin may not be tougher competition in output markets but cheaper and better access to imported intermediates. Any empirical study that ignores input tariffs risks missing the main action.